Extracting Social Determinants of Health from Clinical Notes Using LLMs

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Introduction

- Approximately 80% of factors that affect health outcomes are non-clinical Social Determinants of Health (SDoH)
- SDoH are usually buried in unstructured clinical notes, which makes it hard to extract
- Prior studies [1] on extracting SDoH from clinical notes are limited to a certain type (like BERT), or a certain setting (entity-level vs. note-level), lacking a systematic comparison to inform best practices
- We evaluated state-of-the-art **Causal or Masked LLMs** in a multi-label classification (MLC) for extracting SDoH at the note level

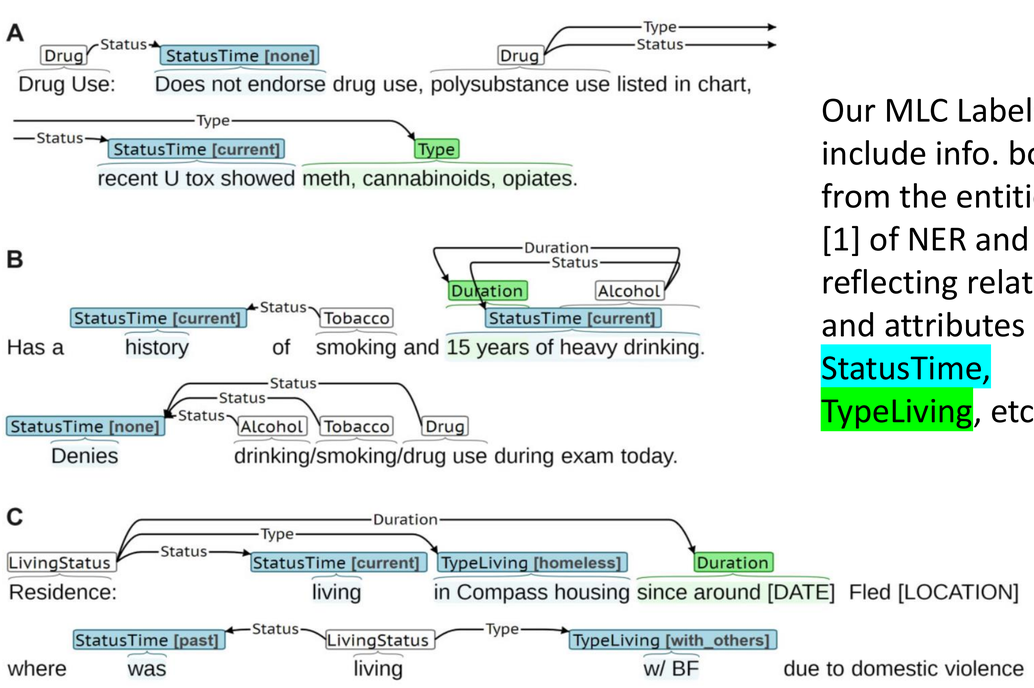
Model

- 6 fine-tuned (FT) Causal LLMs as shown in the right table using Low Rank Adaptation (LoRA) [3] and 4-bit quantization: **Llama-8B, Mistral-7B, DeepSeek-8B, DeepSeek-14B, GPT3.5 Turbo, GPT4o**
- Few-shot GPT-4o: 0-shot, 3-shot, 5-shot
- RoBERTaMLC: RoBERTa FT with label aggregation in pre-processing
- RoBERTaNER: RoBERTa FT with label aggregation in post-processing

Dataset

2022 n2c2 challenge [2] (MIMIC-III; 1,877 notes)

- Example of clinical notes, NER, and our MLC labels



Our MLC Labels include info. both from the entities in [1] of NER and details reflecting relations and attributes like StatusTime, TypeLiving, etc.

Table 1: All Categories and Our MLC Labels (Entities x Note-Level Details)

Category	Entities	Note-level Details
Substance Use	Alcohol Use	None, Current, Past
	Drug Use	None, Current, Past
	Tobacco Use	None, Current, Past
Living Status	Temporal Context	Current, Past, Future
	Living Type	With Family, Homeless, Alone, With Others
Employment	Employment Status	Employed, Unemployed, Retired, On Disability, Student, Homemaker

Result

- GPT-4o (FT) achieves the **highest overall** performance (F1 = 0.900 / 0.963), followed by GPT-3.5 Turbo (0.885 / 0.958)
- **Mistral-7B (0.817 / 0.944) outperforms other open-source models,** including DeepSeek-14B (0.817 / 0.930)
- Few-shot GPT-4o models are competitive with open-source models
- RoBERTaMLC and Llama-8B perform comparably (≈ 0.81 / 0.94)
- **DeepSeek-8B showed the poorest (0.495 / 0.800), failing to match even RoBERTa, mainly due to rare labels (e.g., drug past)**

Table 2: Test Performance (Macro / Micro)

Model	F1	Precision	Recall
RoBERTaNER	0.754 / 0.898	0.753 / 0.905	0.756 / 0.896
RoBERTaMLC	0.812 / 0.941	0.820 / 0.937	0.814 / 0.945
Llama-8B	0.808 / 0.937	0.812 / 0.934	0.807 / 0.940
Mistral-7B	**0.817 / 0.944**	0.829 / 0.940	**0.814 / 0.948**
DeepSeek-8B	**0.495 / 0.800**	0.593 / 0.863	**0.453 / 0.746**
DeepSeek-14B	0.817 / 0.930	0.861 / 0.938	0.805 / 0.921
GPT4o 0-shot	0.581 / 0.687	0.569 / 0.804	0.618 / 0.600
GPT4o 3-shot	0.793 / 0.843	0.774 / 0.790	0.846 / 0.804
GPT4o 5-shot	0.813 / 0.876	0.811 / 0.843	0.851 / 0.912
GPT3.5 Turbo	0.885 / 0.958	0.892 / 0.962	0.884 / 0.953
GPT4o	**0.900 / 0.963**	0.898 / 0.964	**0.903 / 0.963**

Conclusion

- LLMs show promising results for note-level SDoH extraction
- They still cannot excel in labels with implicit meanings (e.g., living with others)
- Future work: Fairness and XAI for rare labels on models **like the DeepSeek series**

Table 3: Micro F1 on Test Rare Labels (Frequency)

FT Model	Drug_past (16)	Living Status_ past (8)	Living Status_ with_others (38)
RoBERTaMLC	0.6667	0.9412	0.7397
Mistral-7B	0.7368	0.8	0.7692
DeepSeek-14B	0.7692	0.3636	0.7143
DeepSeek-8B	0.2857	0	0.2449
Llama-8B	0.6452	0.625	0.7692
GPT3.5 Turbo	0.7586	0.8571	0.7576
GPT4o	0.8387	0.9333	0.7838

Reference

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